MICROBIOLOGY

Blood Flukes (Schistosomes)

LG 1 · Dr Niyan Inaam Muhammed Yousif
GUE BLOCK

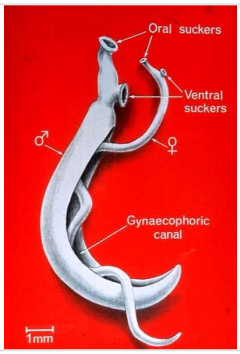
Microbiology GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

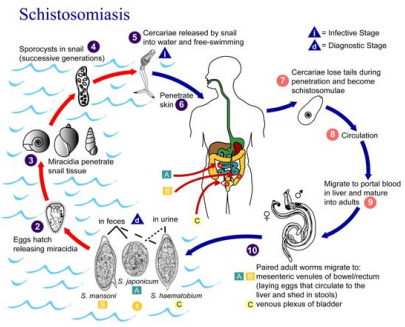
1. Which vein does the **adult worm** of each species live in, and which two species cause **intestinal** (vs **urinary**) schistosomiasis?
2. What is the **infective stage** for humans, and what is the **diagnostic stage** found in stool/urine?
3. Which species' eggs have a **lateral spine**, which have a **terminal spine**, and what long-term complication is S. haematobium uniquely linked to?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

Blood flukes (Schistosomes) — dioecious trematodes (sexes separate) causing **schistosomiasis (bilharziasis)**, a **water-borne disease** affecting millions in Africa, Asia and Latin America. First described by **Theodor Bilharz**, 1851, in the mesenteric veins of an Egyptian in Cairo. >100 million infected with *S. haematobium, S. mansoni* and *S. japonicum*; >200,000 deaths annually. Name = Greek *schisto* (split) + *soma* (body) — the male's body folds into a ventral groove, the **gynecophoric canal**, which holds the female. Male: 10–15 mm, broader/stouter; female: longer, 17–20 mm.



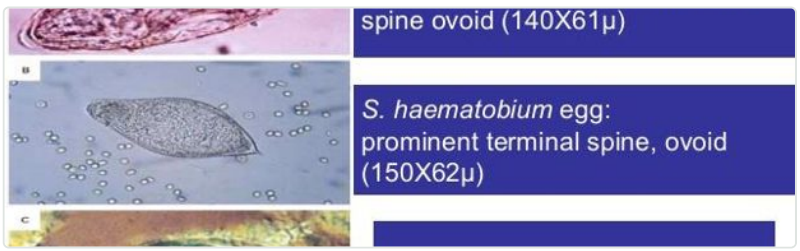
Adult pair — male's gynecophoric canal holding the female



Life cycle — snail (intermediate host) ⇌ human (definitive host)

1 Species discriminator grid

Feature	S. mansoni	S. haematobium	S. japonicum
Adult habitat (vein)	Inferior mesenteric vein (ileocecal)	**Vesical & pelvic plexus** (bladder)	Superior mesenteric vein (ileocecal); also intrahepatic portal venules & hemorrhoidal plexus
Disease type	**Intestinal** schistosomiasis	**Urinary** schistosomiasis	**Intestinal** schistosomiasis
Egg shape/spine	Ellipsoidal, **lateral spine**	Ellipsoidal, **terminal spine**	Round, rudimentary spine (lecture also states "no spine" on the diagnosis slide — **source discrepancy**, confirm with your lecturer)
Intermediate host (snail)	*Biomphalaria*	*Bulinus*	*Oncomelania*
Egg exits via	Feces	**Urine**	Feces



Egg morphology of the 3 key species — the classic diagnostic discriminator



Vesical/pelvic venous plexus — S. haematobium habitat

2 Life cycle — 2 hosts

Hosts	**Definitive host:** humans. **Intermediate host:** fresh-water snails (species-specific, see grid above)
In water	Eggs hatch → release **miracidia** → penetrate snail tissue → **sporocysts** (successive generations in snail) → **cercariae** released into water, free-swimming
Infective stage	**Cercariae** penetrate **unbroken skin** → lose tails during penetration → become **schistosomulae**
In the human	Schistosomulae enter the circulation → migrate to **portal blood in the liver** and mature into adults → paired adults migrate to their species-specific venous site (mesenteric venules or vesical plexus)
Diagnostic stage	**Eggs** — laid in venules, circulate to the liver or are shed directly in stool (S. mansoni/S. japonicum) or urine (S. haematobium)

3 Pathogenicity — 3 clinical stages

1 — Skin penetration & incubation (due to cercaria/ schistosomulum)	Local **cercarial dermatitis** ("**swimmer's itch**") — transient itching + petechial lesions at the cercarial entry site — or general anaphylactic/toxic symptoms. **Acute stage (Katayama fever):** begins shortly after penetration (itching/dermatitis), followed 2–3 weeks later by fever, chills, diarrhea, urticaria, lymphadenopathy and hepatosplenomegaly
2 — Egg deposition (chronic stage)	Most serious pathology — eggs lodge in **liver, spleen, gut wall or bladder wall**. Eggs in the liver induce **granulomas** (response to egg antigens) → **fibrosis**, hepatomegaly, **portal hypertension** → splenomegaly, ascites, drum-like abdominal distention
3 — Tissue proliferation (chronic genitourinary disease)	Generalized hyperplasia/fibrosis of the vesical mucosa — granular **"sandy patch"** appearance. S. haematobium eggs in the bladder wall → granulomas/fibrosis → can progress to **bladder carcinoma (squamous cell)**, presenting with **hematuria**. Also: obstructive hydroureter, anemia (esp. children), secondary bacterial cystitis, bladder calculi (oxalate/uric acid around eggs/clots)

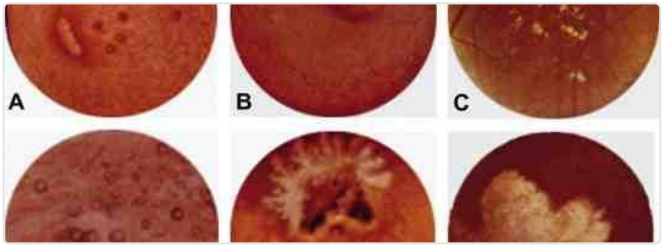
Significant disease may also occur in the **lungs and CNS (spinal cord)**; genital organs may be involved.



Cercarial dermatitis ("swimmer's itch") at the entry site



Hepatosplenic disease — ascites & drum-like abdominal distention from portal hypertension



Cystoscopy: (A) pseudotubercles+ulcer (B) sessile mass (C) sandy patches (D) cystitis cystica (E) malignant ulcer (F) fungating SCC mass

4 Diagnosis, treatment & prevention

Method	Finding
Microscopy	Characteristic **ova in feces or urine** — spine type identifies the species
Histopathology	**Bladder mucosal biopsy** → S. haematobium; **rectal mucosal biopsy** → S. mansoni & S. japonicum. Shows egg clusters in the lamina propria, miracidial nuclei, characteristic spines
Serology	Detection of specific schistosome antigens
Imaging	Ultrasound, CT, X-ray, MRI (e.g. hydroureter/hydronephrosis); cystoscopy
Treatment	**Praziquantel** — treatment of choice for all three species
Prevention	Eradicate intermediate hosts (water snails) · prevent environmental pollution with urine/feces · treat infected persons · avoid swimming/bathing/washing in infected water



Oncomelania hupensis — intermediate host for S. japonicum



Bulinus truncatus — intermediate host for S. haematobium

P Must-remember pearls

- **Vesical/pelvic plexus + eggs in urine** = S. haematobium → the only species causing **urinary** schistosomiasis and the one linked to **bladder squamous cell carcinoma**.
- **Cercariae are infective** (skin penetration); **eggs are diagnostic** (found in stool/urine) — don't mix these up.
- Most pathology comes from the **granulomatous response to eggs**, not the adult worms themselves — this drives both hepatic fibrosis and bladder fibrosis.
- **Katayama fever** = the acute 2–3 week post-penetration syndrome (fever, urticaria, hepatosplenomegaly) — distinct from the early cercarial dermatitis.
- **Praziquantel** treats all three species — one drug, one dose regimen to remember.

Q Test yourself

1. A child who regularly swims in a freshwater lake in an endemic area develops itching and petechial papules at the site of water contact within hours of swimming. Two to three weeks later he develops fever, urticaria and hepatosplenomegaly. What is this later syndrome called?

A. Loeffler syndrome
B. Katayama fever
C. Sandy patch syndrome
D. Swimmer's ear
E. Carrion's disease

2. A 30-year-old man from an endemic area has painless terminal hematuria and a cystoscopy showing granular 'sandy patches' on the bladder mucosa. Which species and habitat best explain this presentation?

A. S. mansoni — inferior mesenteric vein
B. S. japonicum — superior mesenteric vein
C. S. haematobium — vesical and pelvic venous plexus
D. S. mekongi — hepatic vein
E. S. intercalatum — portal vein only

3. Stool microscopy shows an ellipsoidal egg with a prominent lateral spine. Which snail genus would most likely have served as the intermediate host for this infection?

A. Oncomelania
B. Bulinus
C. Biomphalaria
D. Lymnaea
E. Planorbis

4. A patient with longstanding schistosomiasis develops hepatomegaly, splenomegaly and ascites with a distended, drum-like abdomen, but preserved liver synthetic function. What is the primary mechanism?

A. Direct hepatocyte invasion by adult worms
B. Granulomatous reaction to eggs carried to the liver causing fibrosis and portal hypertension
C. Viral co-infection acquired from the same water source
D. Cercarial penetration of the liver capsule
E. Bacterial superinfection of the biliary tree

5. A patient presents years after untreated S. haematobium infection with hematuria. Cystoscopy shows a fungating mass, and biopsy confirms malignancy. What is the most likely histological type, and what is the implicated mechanism?

A. Adenocarcinoma, from chronic urinary stasis alone
B. Squamous cell carcinoma, from chronic granulomatous inflammation and fibrosis driven by egg deposition
C. Transitional cell carcinoma, unrelated to schistosomiasis
D. Lymphoma, from chronic antigen stimulation of lymphoid tissue
E. Renal cell carcinoma, from ascending infection

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — The acute syndrome 2–3 weeks after cercarial penetration — fever, chills, diarrhea, urticaria, lymphadenopathy and hepatosplenomegaly — is Katayama fever, distinct from the earlier cercarial dermatitis (swimmer's itch).

2 — S. haematobium lives in the vesical and pelvic venous plexus draining the bladder, causing urinary schistosomiasis with bladder wall sandy patches and hematuria.

3 — A lateral-spine egg identifies S. mansoni, whose intermediate host snail is Biomphalaria.

4 — Most schistosomiasis pathology is driven by the granulomatous reaction to eggs carried to the liver, causing fibrosis and portal hypertension — hepatocyte function itself is often preserved early on.

5 — Chronic S. haematobium infection causes granulomas and fibrosis of the bladder wall from egg deposition, which is the recognized mechanism linking it to squamous cell carcinoma of the bladder.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam